

- 5 (a) change/loss in kinetic energy = change/gain in electric potential energy B1
 $2 \times \frac{1}{2}mv^2 = q^2 / 4\pi\epsilon_0 r$ C1
 $2 \times \frac{1}{2} \times 2 \times 1.67 \times 10^{-27} \times v^2$
 $= (1.6 \times 10^{-19})^2 / (4\pi \times 8.85 \times 10^{-12} \times 1.1 \times 10^{-14})$ M1
 $v = 2.5 \times 10^6 \text{ m s}^{-1}$ A0 [3]
- (b) $pV = \frac{1}{2}Nm\langle c^2 \rangle$ and $pV = NkT$ C1
 $\frac{1}{2} m\langle c^2 \rangle = \frac{3}{2} kT$ (award 1 mark of first two if $\langle c^2 \rangle$ not used) C1
 $\frac{1}{2} \times 2 \times 1.67 \times 10^{-27} \times (2.5 \times 10^6)^2 = \frac{3}{2} \times 1.38 \times 10^{-23} \times T$ C1
 $T = 5 \times 10^8 \text{ K}$ A1 [4]
- (c) e.g. this is very high temperature
temperature found in stars
(any sensible comment, 1 mark)
(if $T < 10^6 \text{ K}$, should comment that too low for fusion to occur) B1 [1]

- 6 (a) (i) either prevent loss of magnetic flux